Immunoblotting

Risk assessment

- ▷ Wear gloves, safety glasses, lab coat
- ☐ Dispose milk in accordance with local regulations
- ☐ Some jurisdictions prohibit draining dairy products due to their biological oxygen demand

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This is a bench card. Full protocol available online.

›› *Immunoprobing*

- ☐ Incubation tray *or* Resealable plastic bag
- ☐ R0169 TBS with Tween™ 20, 1 L
- ☐ Bovine serum albumin *or* Nonfat dry milk powder ("Blotto")
- ☐ Antibodies, primary and secondary

(1.) Prepare blocking buffer by diluting Blotto to 5% or BSA to 1–5% in 20 mL TBST.

> ***Critical:*** Avoid milk-based blocking buffers such as Blotto for phospho-proteins or streptavidin-based detection. Milk contains phosphatases and biotin, respectively. Phosphatase activity can be quenched with 50 mM sodium fluoride.

(2.) Block the membrane face-up in 5 mL blocking buffer for 30 min at room temperature with agitation. 30 min

(3.) Dilute the primary antibody in blocking buffer (typically 1:1 000 to 1:10 000).

> ***Critical:*** For multiplexed experiments, avoid combining mouse and rat primary antibodies as they are difficult to cross-adsorb. Chicken antibodies tend to non-specifically bind to PVDF and nylon-based membranes, leading to high background.

(4.) Discard blocking buffer. Incubate with diluted primary antibody for 30–60 min at room temperature or overnight at 4 °C with agitation. 30–60 min

(5.) Transfer membrane to a clean tray.

(6.) Wash four times with 10–20 mL TBST for 3 min each with gentle shaking. 15 min

(7.) Dilute secondary antibody in TBST (typically 1:2 000 to 1:10 000) according to vendor instructions.

> ***Critical:*** Beware, secondary anti-mouse antibodies may be subclass-specific for IgG1, IgG2a, or IgG2b. Use highly cross-adsorbed secondaries when multiplexing with antibodies from different species.

(8.) Incubate with secondary antibody at room temperature for 30–60 min with agitation. 30–60 min

(9.) Wash again as before.

A › *Fluorescent detection*

(1.) Specifications for infrared dyes used, for example, with LI-COR Odyssey® Imagers:

- Use IRDye® 800CW or IRDye® 680RD secondary antibodies, at 1:20 000 dilution.
- Use IRDye® 800CW for the lowest-abundance target; it gives the lowest background.
- Rinse the membrane with 1 × TBS to remove residual Tween™ 20.

(2.) Clean the imager surface with distilled water, then 70% ethanol. Dry with a lint-free wipe.

(3.) Scan the membrane using a suitable fluorescence imager.

B › *Luminescent detection*

- ☐ R0118 1 M Tris hydrochloride, 100 mL
- ☐ R0119 100 × Luminol, 10 mL
- ☐ R0120 200 × *p*-Coumaric acid, 10 mL
- ☐ R0121 ECL reagent, 10 mL

(1.) Prepare 10 mL ECL reagent per membrane just before use.

(2.) Drip-dry the membrane and distribute ECL reagent dropwise to cover the surface evenly. Incubate for 1 min at room temperature. 1 min

(3.) Briefly slip the membrane through deionized water and gently drag over the edge of a container to remove excess liquid.

(4.) Expose to film or camera. The signal will last for at least 10 min.

Stripping and reusing membranes

- ☐ R0122 Stripping buffer, 15 mL

(1.) Wash membrane in distilled water for 5 min.

(2.) Strip antibodies using one of the following:

Stripping buffer	**NC**	**PVDF**
0.2 M NaOH		
25 mM Glycine, pH 2.0		
Stripping buffer, pH 6.8	Not recommended	

> ***Critical:*** Do not use high concentration of SDS with nitrocellulose membranes as this can destroy membrane integrity.

(3.) Wash membrane twice in distilled water for 5 min.

(4.) Reblock and reprobe as needed.

Recipe (available online) Troubleshooting (available online) Notes (available online)